

Multi-Season Land Cover Classification & Seasonal Vegetation Analysis

GitHub README — Geospatial Machine Learning Project

Overview

This project combines GIS-based remote sensing analysis, seasonal vegetation assessment, feature engineering, and Random Forest machine learning to analyse land-cover and vegetation dynamics in the Palakkad region of Kerala, India. The analysis uses Sentinel-2 Surface Reflectance Harmonized imagery and Google Earth Engine to derive spectral and vegetation features across different seasons. These features are then used to train and evaluate a Random Forest classifier for five land-cover classes.

Project components: (1) GIS & Seasonal Vegetation Analysis and (2) Random Forest Land-Cover Classification.

Objectives

- Analyse seasonal vegetation behaviour using Sentinel-2 imagery.
- Calculate vegetation, water and moisture indices.
- Identify areas with strong seasonal vegetation variability.
- Build a multi-season feature stack for machine learning.
- Classify land cover using Random Forest.
- Evaluate classification performance using independent validation data.
- Analyse feature importance and class-wise performance.
- Generate a spatial land-cover classification map for the study area.

Study Area

The study focuses on a sub-region of Palakkad District, Kerala, India. The selected area covers approximately 1,270 km² and includes forest areas, agroforestry/homestead vegetation, agricultural and paddy areas, built-up areas, and water bodies.

Data & Technologies

Satellite Data: Sentinel-2 Surface Reflectance Harmonized (COPERNICUS/S2_SR_HARMONIZED).

Land-cover classes: Dense Forest; Agroforestry / Tree-outside-Forest; Seasonal Cropland; Built-up; Water.

Part 1 — GIS & Seasonal Vegetation Analysis

Sentinel-2 imagery was analysed for post-monsoon and pre-monsoon/summer periods. Cloud-contaminated observations were removed and seasonal median composites were generated.

Spectral and Vegetation Indices

- **NDVI:** vegetation greenness and density.
- **EVI:** additional sensitivity to vegetation and dense canopy conditions.
- **NDWI:** supports water and moisture discrimination.
- **NDMI:** characterizes vegetation moisture conditions.

Seasonal Variability

```
ΔNDVI = NDVIprior-monsoon - NDVIpost-monsoon
ΔEVI = EVIprior-monsoon - EVIpost-monsoon
NDVI Range = |NDVIprior-monsoon - NDVIpost-monsoon|
EVI Range = |EVIprior-monsoon - EVIpost-monsoon|
```

Low variability indicates relatively stable vegetation response, while high variability indicates stronger seasonal vegetation response. Seasonal analysis was used as a phenological assessment and the resulting seasonal features were incorporated into the Random Forest feature stack.

Part 2 — Random Forest Land-Cover Classification

The machine-learning feature set combined Sentinel-2 spectral bands, red-edge bands, NDVI, EVI, NDWI, NDMI, seasonal statistics, seasonal variability and texture information. These features provide complementary information about vegetation density, canopy condition, moisture, surface characteristics, seasonal behaviour and spatial texture.

Training Data

- **346** reference polygons
- **62,890** pixel samples
- **47,537** training pixels
- **15,353** independent testing pixels

The data were separated at the polygon level to provide independent validation and reduce the risk of spatial leakage.

Model Evaluation

Metric	Result
Overall Accuracy	94.8%
Kappa Coefficient	0.875

Class	Precision	Recall	F1-Score
Dense Forest	0.980	0.822	0.894
Agroforestry	0.813	0.842	0.827
Seasonal Cropland	0.972	0.974	0.973
Built-up	0.933	0.956	0.944
Water	0.795	0.769	0.782

Feature Importance

Important predictors included summer red-edge bands, NDMI, NDWI, NDVI statistics and seasonal vegetation variability. This demonstrates the value of combining spectral, vegetation, moisture and seasonal information rather than relying only on individual bands.

Classification Challenges

- **Agroforestry vs Dense Forest:** both are tree-dominated and can have similar spectral responses.
- **Agroforestry vs Seasonal Cropland:** mixed pixels and similar vegetation responses can cause confusion.
- **Water and Built-up:** classification can be affected by shadows and other spectrally similar surfaces.

Workflow

```
Sentinel-2 Satellite Imagery
↓
Cloud Masking
↓
Seasonal Composites
↓
Spectral Indices + Seasonal Metrics
↓
Multi-Season Feature Stack
↓
Reference Training Polygons
↓
Training / Validation Split
↓
Random Forest Model
↓
Land-Cover Classification
↓
Accuracy Assessment + Area Statistics
```

Repository Structure

```
land-cover-seasonal-analysis/
■
■■■ README.md
■■■ Gis_Assignment.ipynb
■■■ random_forest_training.ipynb
■■■ requirements.txt
■■■ results/
    ■■ seasonal_analysis/
    ■■ classification/
    ■■ model_evaluation/
```

Notebook Contents

Gis_Assignment.ipynb: Sentinel-2 preparation, seasonal composites, NDVI, EVI, NDWI, NDMI and seasonal vegetation variability.

random_forest_training.ipynb: feature engineering, multi-season feature stack, training data preparation, Random Forest classification, validation, confusion matrix, precision/recall/F1 and feature importance.

Key Results

The project demonstrates an end-to-end remote sensing and geospatial machine-learning workflow. The final reported model achieved **94.8% overall accuracy** and a **0.875 Kappa coefficient** for the Palakkad study area.

Limitations

- Sentinel-2 spatial resolution can produce mixed pixels around small agricultural plots and narrow built-up areas.
- Agroforestry and forest can have similar spectral characteristics.
- Agroforestry and cropland can overlap in mixed landscapes.
- The number of reference polygons for Dense Forest was relatively limited.
- Seasonal composites may not capture all intra-seasonal vegetation dynamics.

Future Improvements

- Integrate Sentinel-1 SAR data and seasonal VV/VH backscatter features.
- Use higher-resolution optical imagery.
- Increase field/reference training samples.
- Improve representation of Dense Forest training data.

- Explore additional temporal observations and alternative machine-learning models.
- Develop a fully interactive web-based geospatial application.

Technologies Used

Python • Google Earth Engine • Sentinel-2 • Remote Sensing • GIS • Random Forest • Scikit-learn • Geospatial Data Analysis • Spectral Index Analysis • Feature Engineering • Jupyter Notebook

Conclusion

This project demonstrates an end-to-end remote sensing and geospatial machine-learning workflow, beginning with Sentinel-2 satellite data and seasonal vegetation analysis and progressing to multi-feature Random Forest land-cover classification. The integration of seasonal vegetation behaviour with spectral and moisture information supports the separation of persistent vegetation, agroforestry, seasonal cropland, built-up areas and water.